



# **Addis Ababa University**

**College of Health Sciences, School of Pharmacy**

**Department of Social and Administrative  
Pharmacy**

**Course title: Pharmacoeconomics (Phar 4211)**

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# Course Objective

**At the end of this course, students will be able to**

- Define economics
- Identifies principles of economics
- Clearly define Pharmacoeconomics and its importance
- Describe the different methods of pharmacoeconomic analysis
- Evaluate pharmacoeconomic studies and effectively use them for decision making.

# 1

## INTRODUCTION TO PRINCIPLES OF ECONOMICS

# Introduction

- ✓ **What is economics?**
- ✓ **Why economics exist as a subject?**

Unlimited human wants Vs Limited resources

- Economics is the study of **how people allocate their limited resources** in an attempt to satisfy their unlimited wants.
- The subject matter of economics **lies on the production, distribution and consumption** of economic goods.
- Economics is a social science that studies **individuals and organizations** engaged in the **production, distribution, and consumption of goods and services**.
- The goal is **to predict economic occurrences and to develop policies** that might prevent or correct such problems as unemployment, inflation, or waste in the economy.

# Use of economics

- To study the behavior of human being in production, distribution, and consumption of goods and services in a world of scarcity to allocate resources
- For resolving or alleviating specific economic problems
- For making business decisions by managing business activities
- For wise policy making
- To prevent future social, political and economic crisis
- To manage our day to day personal lives

# Scarcity

- Scarcity is a **fundamental problem** for every society.
- It refers to **physical condition** where the quantity desired of a particular resource exceeds the quantity available

*Therefore, decisions must be made regarding what to produce, how to produce it, and for whom to produce.*

- **What** to produce involves decisions about **the kinds and quantities** of goods and services to produce.
- **How** to produce requires decisions about **what techniques to use** and how economic resources (or factors of production) are to be combined in producing output
- **For whom** to produce involves **decisions on the distribution of output** among members of a society

# Opportunity cost

- Economic decisions on what, how and for whom to produce involves a choice which **involves sacrifice of the alternative opportunity.**
- **An *opportunity cost* is what is sacrificed to implement an alternative action,** i.e., what is given up to produce or obtain a particular goods or services.
- **Opportunity costs are found in every situation** in which scarcity necessitates decision-making.

Example:

- ✓ Doing trading activities is the opportunity cost for learning in the class
- ✓ If someone was interested to buy house and car at same time and if he bought only house due to shortage of money, the car is the opportunity cost for buying house

# Economic resources

The **economic resources** used to produce goods and services include:

- **Land:** The economy's **natural resources** such as land, trees, and minerals
- **Labor:** The **mental and physical skills** of individuals in a society
- **Capital:** Goods such as **tools, machines, and factories** used in production or to facilitate production.
- **Entrepreneurship:** is a person who **organizes, manages, and assumes** the risks of a firm, **taking a new idea or a new product** and turning it into a successful business

# Production-Possibility Frontier (PPF)

- A production-possibility frontier shows the **maximum number of alternative combinations of goods and services** that a society can produce at a given time when there is full utilization of economic resources and technology.

Eg. If a firm make choices on producing either computer or car as shown below

| Production points | Computer (x) | Car (y) |
|-------------------|--------------|---------|
| A                 | 0            | 10      |
| B                 | 9            | 8       |
| C                 | 20           | 6       |
| D                 | 50           | 4       |
| F                 | 95           | 0       |

## Exercise

- Draw PPF???
- Identify attainable points in the PPF???
- Identify unattainable points in the PPF???

## PPF...

- PPF curve describes three important economic concepts;  
*scarcity, choice and opportunity cost*

Opportunity cost of a good = 
$$\frac{\text{The amount of the next best alternative given up}}{\text{The amount of a good gained}}$$

$$\text{OC of a good} = \frac{\Delta \text{ Good 1}}{\Delta \text{ Good 2}}$$

### **Exercise;**

- Calculate the opportunity cost of producing car at point D, B and A?
- What will be the effect of *technological advancement* and *increasing capital goods* on the PPF curve?

# Branches of economics

Economics has two main branches: 'Micro' and 'Macro' (have been coined by Prof. Ragnar Frisch, 1920's)

## Macroeconomics:

- The study of the behavior of the entire economy and concerned with the behavior of the economy as a whole
- Studies the broad aggregate of economic life such as national output, income, the overall price level, unemployment, and foreign trade.

## Microeconomics:

- Deals with the behavior of individual units of an economy e.g. Consumers, firms, industry and the distribution of production and income among them, ( prices and quantities )(Issues at individual level).

# Significance of Microeconomics

- In the efficient management of the limited resources
- To understand the working of free enterprise economy
- Explain the gains from international trade, balance of payments disequilibrium and determination of foreign exchange rate.
- Explains how through market mechanism goods and services produced in the community are distributed.
- Formulation of economic policies
- Is the basis of welfare economics.
- For constructing economic models

# Limitations of Microeconomics

- It is not capable of explaining the functioning of an economy as a whole.
- It assumes full employment; which is rare in real life.
- It cannot be used for solving the problem relating to public finance, monetary and fiscal policy etc.

# Economic Value

Distinguish two alternative approaches

- ❑ **Deontological view:** nature has value in its own right
- ❑ **Anthropocentric view:** nature has value only to the extent that (some) people value it.
  - It is the latter view that justifies the use of economics when discussing the value of the environment.
  - Economics is **inherently anthropocentric in its focus**: it deals with how and why people behave.
  - The meaning of value in economics is controversial by different scholars. Most agree on "*Intrinsic worth or price of a good or service*".
  - Value as an **indication of relative ranking** e.g. a human life is more important (valuable) than the life of a horse; a horse is more valuable than a donkey.
  - Value as **a measure of trade of** (value in exchange): The value of X in terms of Y (eg. Monetary exchange)

Value...

## Value and satisfaction

- Consumers choose among the products **a particular product** that give them maximum value and satisfaction ( that deliver the maximum value)

## Customer value (CV)

- Is the **difference** between the value the customers gains from owning and using a product or service and the cost of obtaining the product/service.
- In short:  $CV = CG - CC$ 
  - CV= customer value
  - CG= customer gain
  - CC= customer cost

# Utility

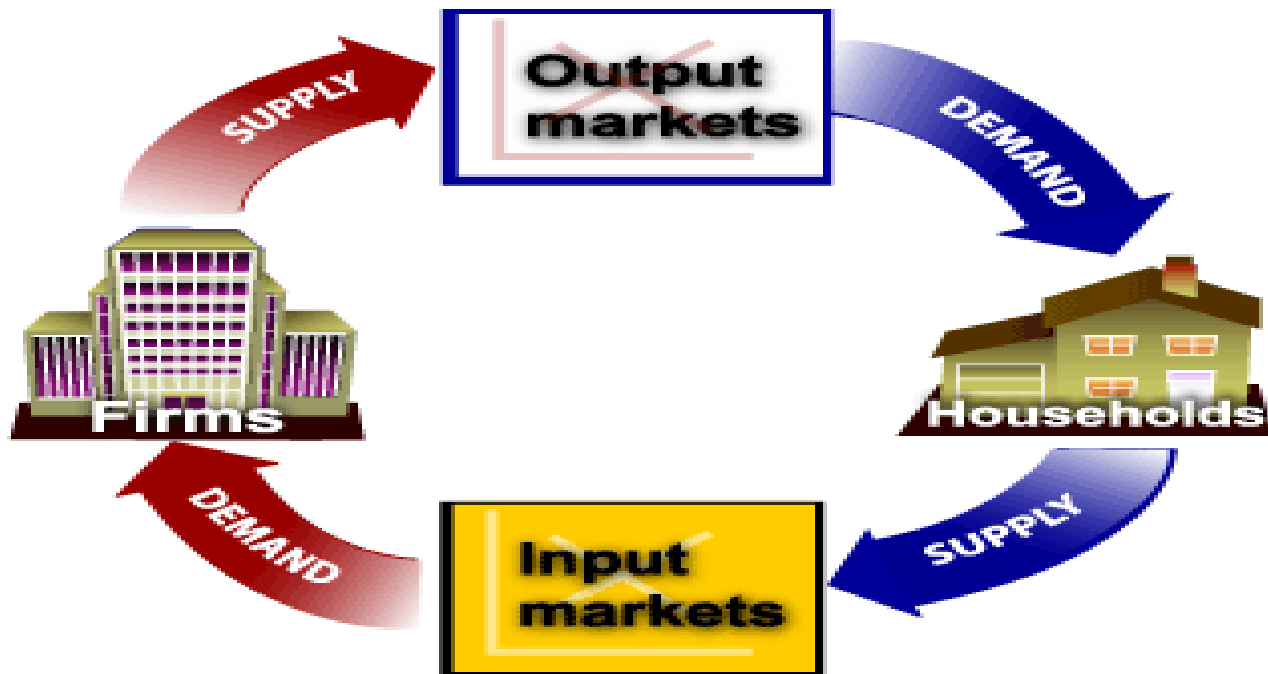
- **Utility:** means **happiness**, or **well-being**. It is a numerical indicator of a person's preferences in which higher levels of utility indicate a greater preference.
- **Total utility:** is the **total amount of happiness experienced by the individual**. Economists usually assume that *it is the goal of the individual to maximize her/his total utility*.
- **Marginal utility:** is **the increase in total utility** when one more unit of an activity is undertaken (extra satisfaction).  
**Example:** consider the activity "consuming potato chips."

| Potato chips eaten<br>(in pack) | Marginal chip<br>utility | Total chip utility |
|---------------------------------|--------------------------|--------------------|
| 0                               | --                       | 0                  |
| 1                               | 50                       | 50                 |
| 2                               | 40                       | 90                 |
| 3                               | 30                       | 120                |
| 4                               | 15                       | 135                |
| 5                               | 5                        | 140                |

## Diminishing marginal utility

- Note from the above table that all 5 of the chips add to the individual's total utility (happiness).
- However, the 2nd chip adds less marginal utility than the 1st chip; the 3rd chip adds less utility than the 2nd chip, etc.
- The condition in which marginal utility falls as consumption of a good rises is known *as the law of diminishing marginal utility*

# Demand and Supply



- The ***circular flow of economic activity*** shows the connections between firms and households in input and output markets.
- A ***firm*** is an organization that transforms resources (inputs) into products (outputs).
- ***Households*** are the consuming units in an economy.

## Demand..

Before we define demand, let us see two terms

### Needs:

- ❖ A state of **felt deprivation for basic items** such as food, clothing, pharmaceuticals and complex needs such as the need for belonging
- ❖ Basic human requirements
- ❖ These needs are not created by marketers. **They are basic part of the human make up**

### Wants

- ❖ Are effective **desires** for specific satisfiers of needs
- ❖ Needs becomes wants when they are directed to specific objectives that might satisfy the need
- ❖ Human wants are **continually shaped and reshaped** by culture, social forces and institutions including churches, schools, families....

## Demand

- Is **ability and willingness to buy** the Product /service  
(***Demand=Need + willingness+ ability***)
- **It should be** supported by purchasing power.
- Therefore, demand is the amount of a good that consumers are willing and able to buy at a given price
- Thus **economists** must therefore measure not only how many people want products but also how many would actually be willing and able to buy it.
- **Quantity demanded:** is the amount (number of units) of a product that a household would buy in a given time period if it could buy all it wanted **at the current market price.**

## Demand....

- **Demand schedule:** is a table that specifies the units of a good or service that the individual is willing and able to purchase **at alternative prices** during a given period of time.
- **Demand curve:** is a graph illustrating how much of a given product a household would be willing to buy at different prices (**graphical representation of demand schedule**)
- **Demand equation:** if all factors except price remain unchanged, the demand function can be expressed:

$$Q_d = f(P)$$

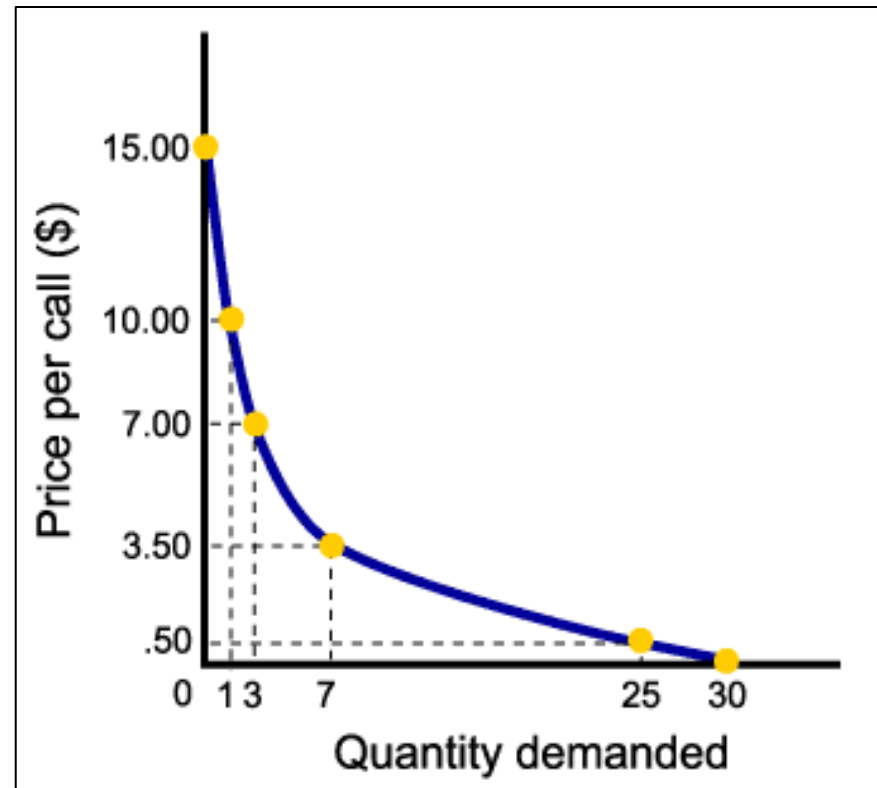
- The **law of demand** states that “ *there is a negative or inverse relationship between the quantity of a good demanded and its price*”.
- Meaning demand **curves slopes downward**.
- More units are purchased at lower prices because of a **substitution effect and an income effect**.

## Demand...

### ANNA'S DEMAND SCHEDULE FOR TELEPHONE CALLS

| PRICE<br>(PER<br>CALL) | QUANTITY<br>DEMANDED<br>(CALLS PER<br>MONTH) |
|------------------------|----------------------------------------------|
| \$ 0                   | 30                                           |
| 0.50                   | 25                                           |
| 3.50                   | 7                                            |
| 7.00                   | 3                                            |
| 10.00                  | 1                                            |
| 15.00                  | 0                                            |

**Demand schedule**



**Demand curve**

- Demand curves intersect the quantity ( $X$ )-axis, because of **time limitations and diminishing marginal utility**.
- Demand curves intersect the ( $Y$ )-axis, **as a result of limited incomes and wealth**.

Demand...

## Why Negative slope?

- The negative relationship between price and quantity occurs due to the following three reasons
  - **Substitution effect:** as the relative price of the good rises, customers substitute the cheaper good as alternative choice.
  - **Income effect:** as the **price of the commodity falls**, the purchasing power of customers' limited income increases and vice versa.
  - **The law of diminishing marginal utility:** as the customer increase the consumption of good **beyond a certain level**, the marginal utility of a good tends to decrease.

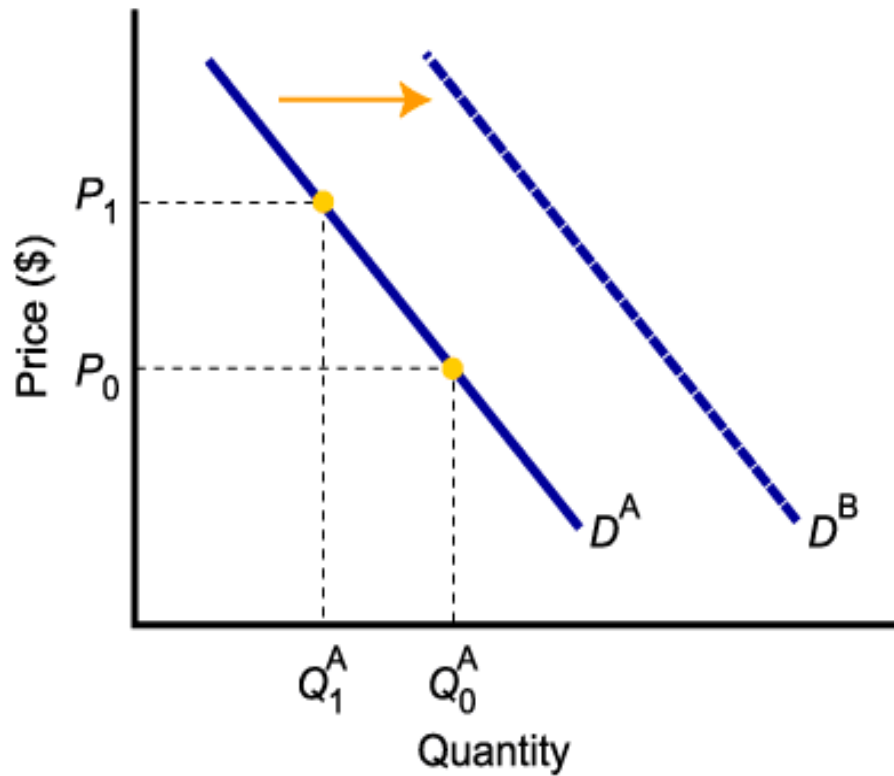
# Determinants of demand

- There are different factors affecting customer demand to goods and services. These factors may cause a change in quantity demanded or demand
- A change in  $Q_d$  is expressed as a movement from one price-quantity combination to another point on a fixed curve whereas a change in Demand (D) means, a change in the demand schedule or shift of the demand curve to the new location.

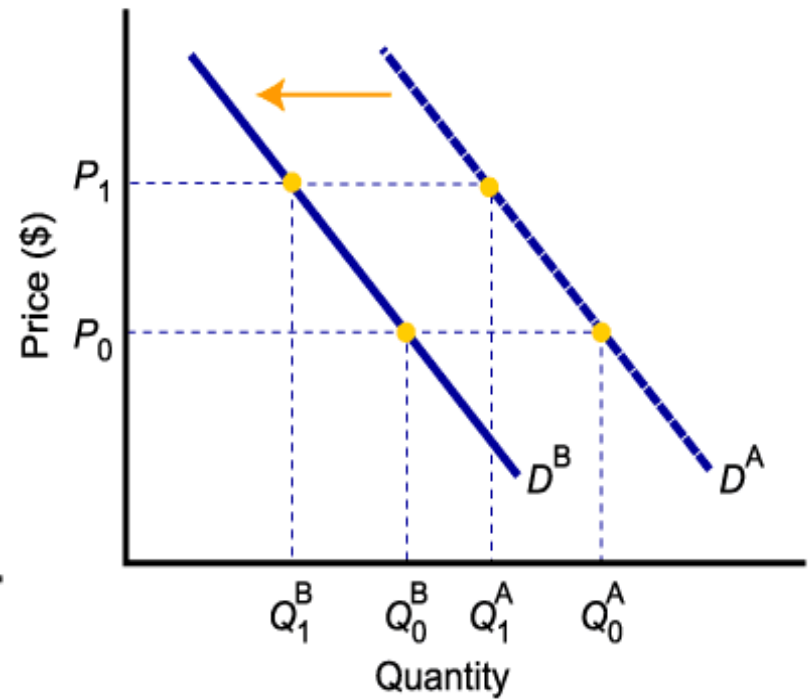
The common determinants are:

- Price of the product
- Taste or preferences of customers
- Income of the consumers
- Price of related goods
- Consumers' expectation of income and price in the future
- Number of buyers in the market

# Determinants...



Increase in demand



Decrease in demand

## Determinants...

### 1. Price

- ❑ A change in price causes a change in quantity demanded represented as a movement along a fixed demand curve.
- ❑ As price increases, the Qd decreases, other things remain constant and vice versa

### 2. Taste or preferences of customers

- The **favourable change in consumers' taste** causes an increase in demand which is represented by a rightward shift of the demand curve

Eg. Fridge having a heater

- The **non favourable change in consumers' taste** due to a scientific report about the a negative impact of a product, or other customers rumour on quality decreases the demand. The demand curve shifts to the left

eg. Drug, Food .....

### 3. Change income of customers

- The change in demand due to the change in income depends on the nature of the product.
- ***Normal Goods:*** are goods for which demand goes up when income is higher and for which demand goes down when income is lower eg. Milk, leather products
- ***Inferior Goods:*** are goods for which demand falls when income rises and vice versa. eg. Sherro wet, Cabbage

# Determinants...

## 4. Effect of changes in prices of related goods

Two goods can be considered as **related goods** if they are a **substitute for** or **complement** to each other.

- ***Substitute goods***: are goods that can serve as replacements for one another; **when the price of one increases, demand for the other goes up.** *Perfect substitutes* are identical products. Eg. Coca and pepsi
- ***Complement goods***: are goods that “go together”; a decrease in the price of one results in an increase in demand for the other, and vice versa. Eg. Oil and onion

# Determinants....

## 5. Effect of changes in consumers' expectation of price and income

- If consumers **expect higher price** of a product in the future, they would buy today to avoid unnecessary expenditure. This results an increase in the current demand
- Consumers' expectation of future income also affects current demand
  - ✓ expectation of higher income will induce consumers to buy more of a normal goods but less of an inferior goods
  - ✓ expectation of lower income will induce consumers to buy more of a inferior goods but less of an normal goods

# Determinants....

## **6. Effect of change in number of buyers of a product**

- When the number of buyers increases at specific product price, the demand will be increased

# Exercise

## Given

| Market point | Price/Kg | Qd/Week |
|--------------|----------|---------|
| A            | 6        | 10      |
| B            | 5        | 20      |
| C            | 4        | 30      |
| D            | 3        | 40      |
| E            | 2        | 50      |
| F            | 1        | 60      |

- Calculate the slope of a function
- Derive the demand function in the form of  $Q_d = f(P)$
- Draw the demand curve

# Supply

- **Supply:** the amounts of a good producer are willing and able to sell at a given price.
- **Quantity supplied** represents the number of units of a product that a firm would be willing and able to offer for sale at a particular price during a given time period

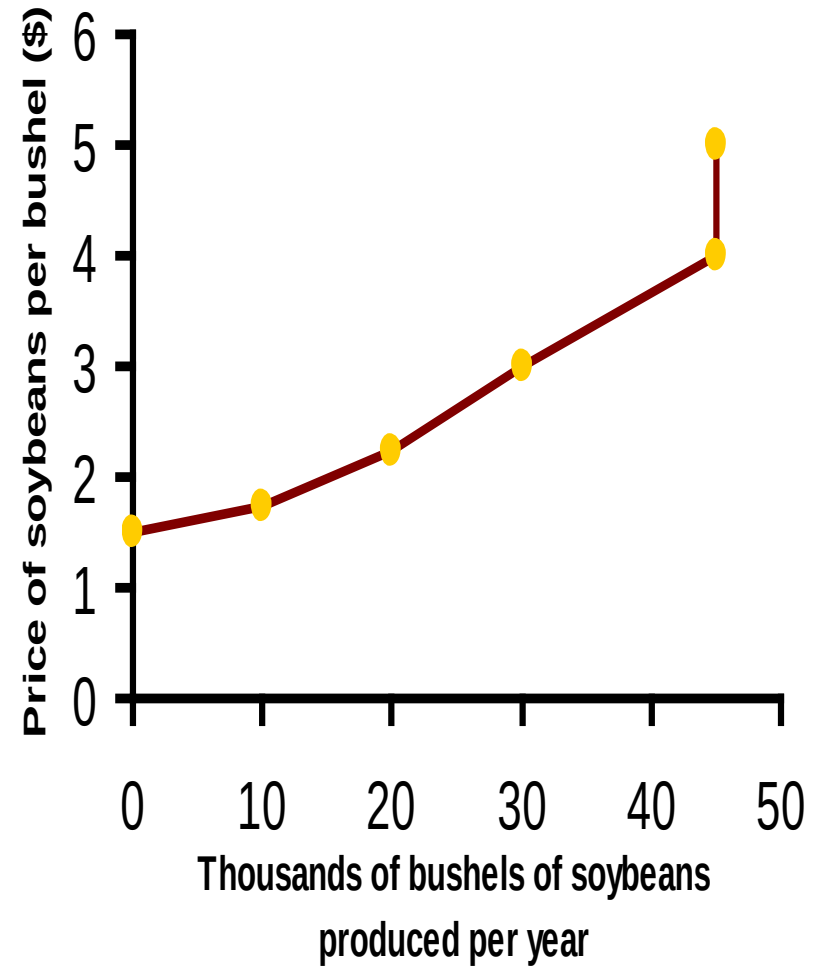
It can be expressed:

- **supply schedule:** is a table showing how much of a product firms will supply at different price.
- **supply curve** is a graph illustrating how much of a product a firm will supply at different prices (graphical representation of supply schedule)
- **Supply Equation:**  $Q_s = F(P)$ 
  - The **law of supply** states that " *there is a positive relationship between price and quantity of a good supplied.*"
  - This means that supply curves typically have a positive slope.

# Supply...

## CLARENCE BROWN'S SUPPLY SCHEDULE FOR SOYBEANS

| PRICE<br>(PER<br>BUSHEL) | QUANTITY<br>SUPPLIED<br>(THOUSANDS<br>OF BUSHELS<br>PER YEAR) |
|--------------------------|---------------------------------------------------------------|
| \$ 2                     | 0                                                             |
| 1.75                     | 10                                                            |
| 2.25                     | 20                                                            |
| 3.00                     | 30                                                            |
| 4.00                     | 45                                                            |
| 5.00                     | 45                                                            |



# Determinants of supply

The following factors may change the entire supply schedule or a shift in the supply curve.

1. Effect of change in input price
2. Effect of change in technology
3. Effect of change in price of related goods
4. Effect of change in sellers' price expectation
5. Effect of changes in taxes
6. Effect of changes in subsidies
7. Effect of changes in number of sellers
8. Effect of change in weather condition

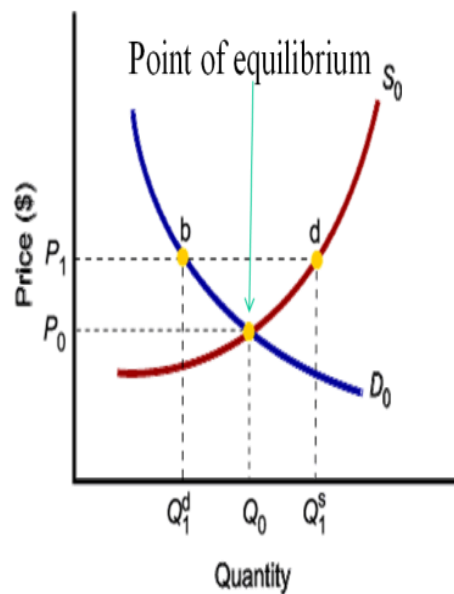
# exercise

- If one firm's supply rate of product X is given as;  
 $Q_s = 30p$ 
  - How much quantity will the firm supply at price 10 birr
  - Calculate the slope
  - Draw the supply curve
  - What will happen to the supply curve if higher tax is incurred to the firm?

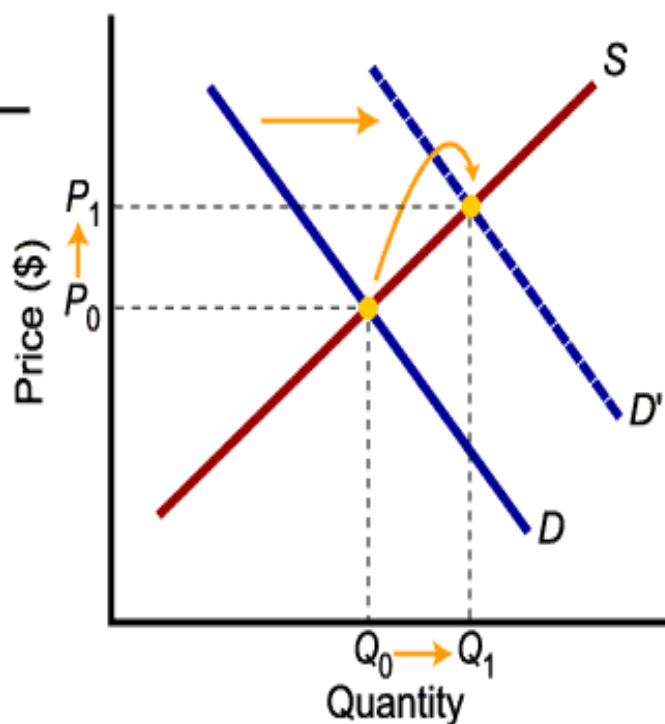
# Market Equilibrium

- The operation of the market depends on the interaction between **buyers and sellers**.
- Equilibrium occurs at the **intersection** of the market supply and market demand curves.
- At this intersection, quantity demanded equals quantity supplied, i.e., the quantity that individuals are willing to purchase exactly equals the quantity producers are willing to supply.
- Equilibrium price (**Market clearing price**): a price at which market demand for a product equals market supply.
- Equilibrium Quantity: the quantity which is actually bought and sold at the equilibrium price.

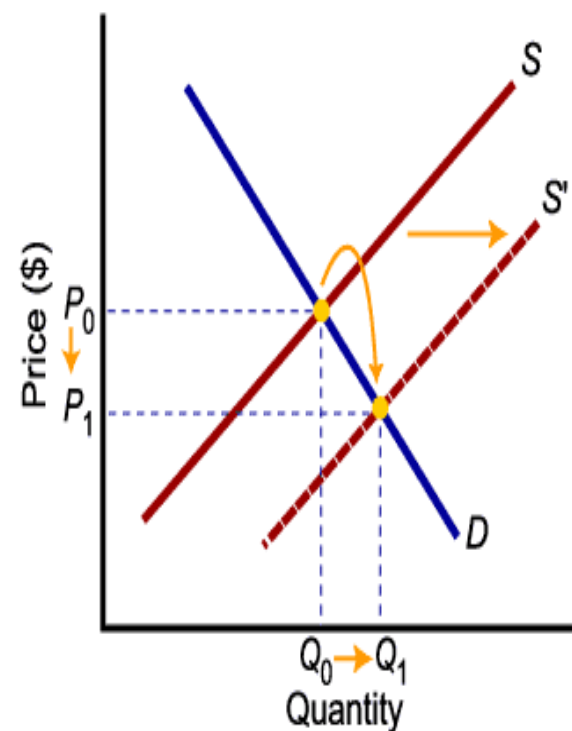
# Market equilibrium cont...



Increase in Demand



Increase in supply



# Market equ...

- At equilibrium price, **there is neither shortage nor surplus**

**Shortage:** the excess of demand over supply

**Surplus:** the excess of supply over demand

## Example:

Consider that there are 200 identical buyers of product X in a market with an individual demand function of  $Q_{dx}=8-P_x$ , other things remain constant. There are also 100 identical sellers of product X with an individual supply function of  $Q_{sx}=2p_x$ , other things remain constant

- Calculate equilibrium price
- Calculate equilibrium quantity
- Draw the supply and demand curve in the same plane

# Assignment

What will happen to the equilibrium price, the equilibrium quantity, and the curve

1. When demand increase while supply is constant
2. When demand decreases while supply constant
3. When supply increases while demand constant
4. When supply decreases while demand constant
5. When demand increases while supply increases
6. When demand decreases while supply increases
7. When demand increases while supply decreases
8. When demand decreases while supply decreases

# Elasticity of Demand and Supply

- Elasticity is a measure of responsiveness of a dependent variables to changes in an independent variable.
- In economics, elasticity measures the responsiveness and sensitivity of consumers or sellers to changes in price or other variables.
- It provides a way of measuring how sensitive demand or supply is to factors such as a change in price.
- **The elasticity of demand (*ED*)** measures the percentage change in the quantity demanded per period of a commodity as a result of a given percentage in its price, ceteris paribus (assuming income and prices of other goods remains constant).
- The formula is

$$ED = \frac{\text{percentage change in the quantity demanded}}{\text{percentage change in price}}$$

## Elasticity...

- Ed is usually a negative number. This is because of the law of demand. To avoid negative, we usually take the absolute value of Ed.
- If Ed is greater than 1, the demand is considered as elastic. This means the customers are sensitive to price change to a product
- If Ed is less than 1, the demand is considered as inelastic. This means the customers are less sensitive to price change to a product
- If Ed is equal to 1, the demand is considered as unitary elastic. This means that the change in quantity demand and price goes proportionally.

## Elasticity..

### Exceptional types of demand curves

- Perfectly elastic demand (  $E_d = \infty$  )
- Perfectly inelastic demand ( $E_d = 0$ )
- **The *elasticity of supply* ( $ES$ )** measures the percentage change in the quantity supplied of a commodity as a result of a given percentage change in its price.

$$ES = \frac{\text{percentage change in the quantity supplied}}{\text{Percentage change in price}}$$

# Exercises

**A.** Suppose that the price of a treatment rose from 20 Birr an hour to 22. At 20 Birr an hour, consumers were buying 1,000 treatments per week and spending 20,000 Birr. After the price rise they bought 950 a week, but their spending had risen to 20,900 Birr. Calculate  $E_D$  and interpret it???

# MARKET STRUCTURE

## What is market?

- Is a condition in which, **potential buyers and sellers come up together** and communicate to buy and sell with one another. It consists of people willing to buy and sell.
- It is used to **describe any process of exchange** between buyers and sellers.
- Formally, a market can be defined as **any set of arrangements** that allows buyers and sellers to communicate and thus arrange exchange of goods, services or resources.

# Market...

- A **free market** is where such exchange occurs without interference from the government.
- **Information** is a vital ingredient for any market.
- Both buyers and sellers need to have access to sufficient information to allow them to make rational decisions.

# Market structure and price

- Market structure refers to the set of industry characteristics that affect the extent of rivalry in the market, and ultimately affects market performance related to pricing and output.

## Elements of market structure:

- *Number and size* of existing and potential entrant sellers & buyers.
- The extent of *entry/exit costs* and other difficulties associated with market entry or exit.
- The extent to which products are *similar or different (product differentiation)* across firms
- The extent of *economics of scale* in production
- The extent to which costs are *similar or different* across firms
- The extent of *transaction costs*, and the extent of *travel or information* costs on the part of buyers.

# 1. Pure/perfectly competitive market

- It is a **hypothetical market** form in which **no producer or consumer** has the market power to influence prices in the market.
- The analysis of perfectly competitive markets **provides the foundation** of the theory of supply and demand.
- There is **no government restriction** on the entrance and exit of sellers and buyers.

*A perfectly competitive market must have the following characteristics*

- Many buyers and many sellers :buyers will not have any power to influence price
- A homogenous product
- Free entry and exit
- Sufficient knowledge

## Pure market...

In general, perfectly competitive markets are likely exhibit the following features

- Lower prices
- Suppliers face highly elastic demand
- Low barriers to entry
- Lower profits
- Economic efficiency

## 2. Monopolistic competition

- Is a market situation in which there are many independent buyers and many independent sellers but in which competition is imperfect because of product differentiation, geographical fragmentation of the market or some similar condition.
- It is a cross between the two extremes of perfect competition and monopoly.

## *Monopolistic comp....*

### *characteristics of monopolistic competition market*

- Has many firms/many competing products
- Faces a down warded demand curve
- Small market share
- Independent actions
- No collusion
- Easy entry and exit
- Product differentiation: product quality , services , location and advertizing and packaging

### 3. Oligopoly

- Is a market situation in which, **a few firms** are participated in the market and, **sale either homogenous or differentiated products**.
- It is **ranging between** pure monopoly and monopolistic competition market structures.

*Oligopoly has the following characteristics*

- Homogeneous or differentiated products
- Mutual interdependence
- Collusion
- Much or limited price control
- Barriers to entry: Natural (economics of scale), artificial barriers (government rules)

## 4. Monopoly

- It is a market structure exists when a **single firm** is the sole producer of a product.
- It is the extreme **opposite** of perfectly competitive market.

*It has the following features*

- No other firm competing with it.
- There is no close substitute of the goods produced and sold by the monopolists
- Buyers purchase at whatever price sated
- Price maker: he can control the price and decide its own price
- Blocked entry: there are barriers to entry

Example:- electricity, telecommunication, etc.

*What is the elasticity of the demand curve?*

# Setting Price

- Pricing can be a **means of competing** not only to take customers of rivals but **to prevent competition** from rivals.
- How a business approaches its pricing strategies **depends on the market structure** it operates

**Price takers:** **have little or no control over** the price they charge.

- In Perfect Competition
  - Firms have to take the price set by the market
  - Large number of sellers – each has small market share and therefore no control over the market.

Example: agricultural products

# Price...

- **Price Leadership:** Dominant firm sets price, the rest have to take this price.
- Occurs where a **dominant firm in an industry in which products are good substitutes.**
- The firm is able to set price and other firms on account of their smaller size will follow the price.
- **Price Fixing:** where firms fix prices at levels above equilibrium **on account of their market power** or through selling/distribution arrangements generally termed collusion. Eg. In Oligopoly market

# Market failure

**Definition:** It is a condition where the market mechanism fails to allocate resources efficiently.

- **Social Efficiency** = where external costs and benefits are accounted for
- **Allocative Efficiency** = where society produces goods and services at minimum cost that are wanted by consumers.

**NB:** Resources cannot be readjusted to make one consumer better off without making another worse off

- **Technical Efficiency** = production of goods and services using the minimum amount of resources
- **Productive Efficiency** = production of goods and services at lowest factor cost

## Market Failure...

- **Market Failure occurs where:**
  - Knowledge is not perfect - ignorance
  - Goods are differentiated
  - Resource immobility
  - Market power
  - Services/goods would or could not be provided in sufficient quantity by the market
  - Existence of external costs and benefits
  - Inequality exists

# Imperfect Knowledge:

- Consumers do not have adequate technical knowledge
- Advertising can mislead or miss-inform customers
- Producers unaware of all opportunities
- Producers cannot accurately measure productivity
- Decisions often based on past experience rather than future knowledge

# Goods/Services are differentiated

- Branding
- Designer labels - they cost three times as much but are they three times the quality?
- Technology—lack of understanding of the impact
- Labelling and product information
- Packaging materials
- Transport scheme

# Resource Immobility

- Factors are not fully mobile
- Labour immobility– geographical and occupational
- Capital immobility– what else can we use our capital?
- Land–cannot be moved to where it might be needed

# Market Power:

- Existence of monopolies and oligopolies
- Collusion
- Price fixing
- Abnormal profits
- Rigging of markets
- Barriers to entry

# Measures to correct market failure

- State provision
- Extension of property rights
- Taxation
- Subsidies
- Regulation
- Prohibition
- Positive discrimination
- Redistribution of income

## *Exercises*

1. What is the difference b/n pure competition and monopolistic competition?
2. Differentiate oligopoly and monopoly markets?
3. What type of demand curve does each market structure exhibit?
4. List at least four features for pure competitive market?

# INTRODUCTION TO PHARMACOECONOMICS

# Presentation outline

- Introduction
- Economic Evaluation
- Importance of pharmacoeconomic evaluations
- Pharmacoeconomic evaluation methods
- Steps of PE evaluation
- Summary

# Objectives of the presentation

- Define Pharmacoeconomics
- To introduce terminologies related with PE evaluation
- Discuss rationale for its use
- To describe pharamcoeconomic evaluation
- Explain the importance and applications of PE
- Identify the beneficiaries from PE evaluation
- Identify the basic PE evaluation techniques
- Ascertain the steps in PE evaluation

# Introduction

## Fundamental Economic Problem:

**Scarcity of resources** Vs **Unlimited human wants**

**Economics:** The study of the appropriate allocation of limited (scarce) resources among alternative uses to satisfy unlimited wants of human being.

# Why Economics for Health?

- Because resources are limited, health economists are concerned with determining **what medical services to produce, how they should be produced, and who should receive them.**
- As we will see in this course, **the tools of economics can be applied** to the health care sector to derive valuable insights about our health care system.

# Context of PE

- In the health service → limited budget

increased spending on drugs takes resources away from other areas of health care.

# Context of PE...

- to provide **high-quality patient care** in the **least expensive way**, clinicians have developed strategies for containing costs
- **most of these strategies focus solely on determining the least expensive alternative**
- rather than the alternative that represents **the best value for the money**

# What is pharmacoeconomics?

- Is a multidisciplinary concept
- Encompasses **pharmacy** and **economics**
- The **study of the financial aspects** of pharmaceutical care
- Is thus the **science of applying economic theory** to pharmaceutical healthcare
- Scarce pharmaceutical resources are put to the best possible use and implemented to achieve the best cost to effect benefits.
- **Pharmacoeconomics:** It is the description and analysis of the **costs** and **consequences of pharmaceutical products** and services and their impact on individuals, health care systems and the society

# PE...

- *Pharmacoeconomics*
  - identifies,
  - measures, and
  - compares
- the costs and consequences of pharmaceutical products and services
- to determine which alternative yields the optimal outcome per dollar/pound/birr spent

# Economic evaluation

**Economic evaluation:** involves the **identification, measurement, valuation**, and then comparison of the **costs (inputs)** and **benefits (outcomes)** of two or more alternative programs or activities (WHO, 2000).

- The comparative analysis of **alternative courses of action in terms of both their costs and consequences.**
- Economic evaluation of health care programmes **aims** to aid decision-makers to allocate health care resources appropriately, setting priorities and moulding health policy.

## Economic evaluation...

**Pharmacoeconomic evaluation:** it is part of economic evaluation which involves the identification, measurement, valuation, and comparison of the costs of pharmaceutical products and services (inputs) and benefits (outcomes) of two or more alternative pharmaceutical products or services.

### Pharmacoeconomic equation

Costs (\$\$) → Rx → outcomes

- **It helps to assess** whether the scarce resources are being spent wisely on pharmaceutical products and services.

# PEE...

- Pharmacoeconomic evaluation is becoming
  - **'the fourth hurdle'** -- newly developed medicines are supposed to pass after the traditional three requirements.
  - **the efficacy**  
is there a **beneficial effect** from the therapy?
  - **safety** (how acceptable are **the side effects**?) and
  - **the quality** (standard manufacturing?)

# Why PE evaluations become important?

## 1. High cost of drugs

- In 2004, US spent almost \$2 trillion on health care costs (10%, or \$200 billion was spent on pharmaceuticals)
- Brazil, about 8% of GDP spent on health care
- Developing countries (spent 40% of health care budget)
- **Ethiopia:** Total annual Expenditure on Drugs
  - Birr 1,499,989,690 in 2003/04 E.C
  - Birr 2,439,186,538 in 2005/06 E.C from all sources

2. Advances in medical technology and drugs are dramatically improving patient care. But, these improvements are costly.

## Why...

- 3.** High cost of research and development (R&D) for new treatments (\$1-2 billion to bring a new drug)
- 4.** Increased market competition among brand and generic products
  - Generic substitution (generic price –20-70% lower than patented drug price)
- 5.** Pharmaceutical sector becomes the most promising business opportunity for investors
- 6.** Increased cost containment efforts
  - Changes in government reimbursement of health care providers
  - Private insurers are exercising more control over patient care

# Why...

- 7.** Accessibility of essential drugs is still a global problem
  - More than 50% of the population in the African Region still have no regular access to EMs (WHO, 2002).
- 8.** Increase irrational drug use practice
  - The WHO estimates that up to 75% of antibiotics are prescribed irrationally and
  - Globally, only 50% of patients take their medicines correctly (WHO, 2002).
- 9.** Antimicrobial Resistance becomes a public health crisis

# Why...

**10.** PE evaluations studies become an integral part for developing:

- Formularies
  - STGs and
  - pharmaceutical care programs
- Applying PE evaluation for comparing pharmaceutical services and treatment options is an accepted and valuable tool in **developing countries** (Mori *et al.*,2013).

# Cases

## Case 1:

- Is it worthwhile to pay \$15,000 per patient per year for a new drug to treat **Multiple Sclerosis** when its benefit is to decrease the frequency of relapses by one-third? Without this drug patients with moderate MS will relapse about once every 2 years. The drug must be injected twice a week and 50% of patients experience debilitating flu-like symptoms that may last several months or more.

# Cases...

## **Case 2:**

Is it worthwhile for a provincial Ministry of Health to reimburse pharmacists at a rate of 300 birr per patient for a disease management program in congestive heart failure?

### Decision

Depend on the countries status and the service delivered out come

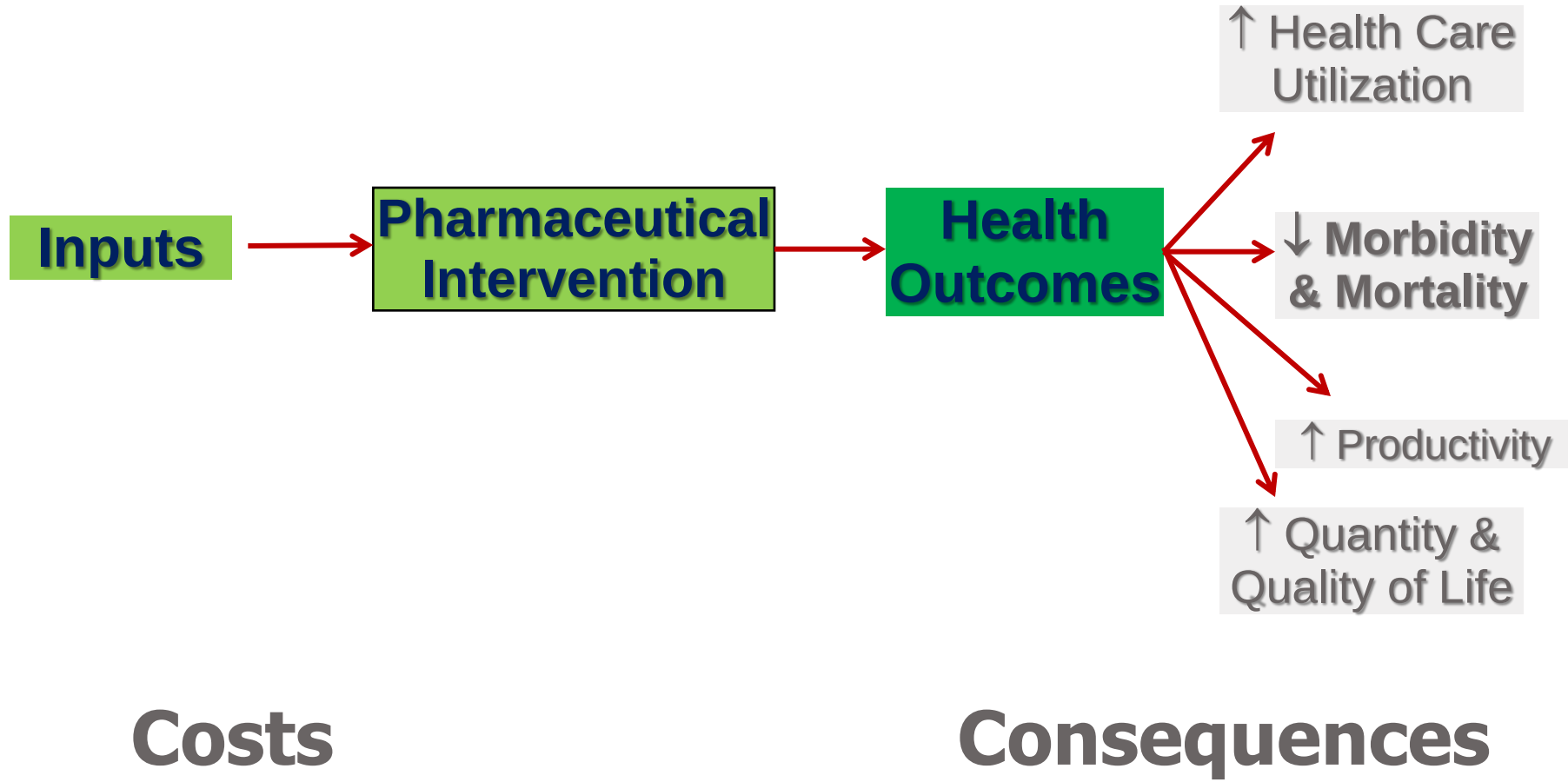
## **Case 3:**

Is it worthwhile to pay \$56,000 per treatment course for a 2<sup>nd</sup> line cancer chemotherapy drug that delays tumor growth but is not known to extend lives?

# Who will be benefited from PE evaluation?

- Government /health policy makers
- Drug regulatory authorities
- Health professionals
- Health insurance companies
- Pharmaceutical Industries
- Pharmaceutical sectors
- Health institutions
- Patients
- Society

# Cont...



# Consequences

A measure of the effects, outputs or outcomes of program or drug therapy of interest

**Table: Outcomes of pharmaceutical care**

| Outcome    | Description                                                               |
|------------|---------------------------------------------------------------------------|
| Clinical   | Eradication of disease and physical survival.                             |
| Humanistic | Quality of life, including physical, emotional and social well-being.     |
| Economic   | Cost of treatment options, productivity and the global effect on society. |

# PE Evaluation methods

There are **four common evaluation techniques** in the area of pharmaceutical care:

1. Cost minimization analysis (CMA)
2. Cost Effectiveness Analysis (CEA)
3. Cost Utility Analysis (CUA)
4. Cost Benefit Analysis (CBA)

Differentiated by **how the health consequences of the treatment are valued.**

# Cost-Of-Illness Evaluation

- Identifies and estimates the **overall cost of a particular disease** for a defined population
- Referred to as *burden of illness*
- Measures the **direct** and **indirect costs** attributable to a specific disease
- Establishes a **baseline for comparison**

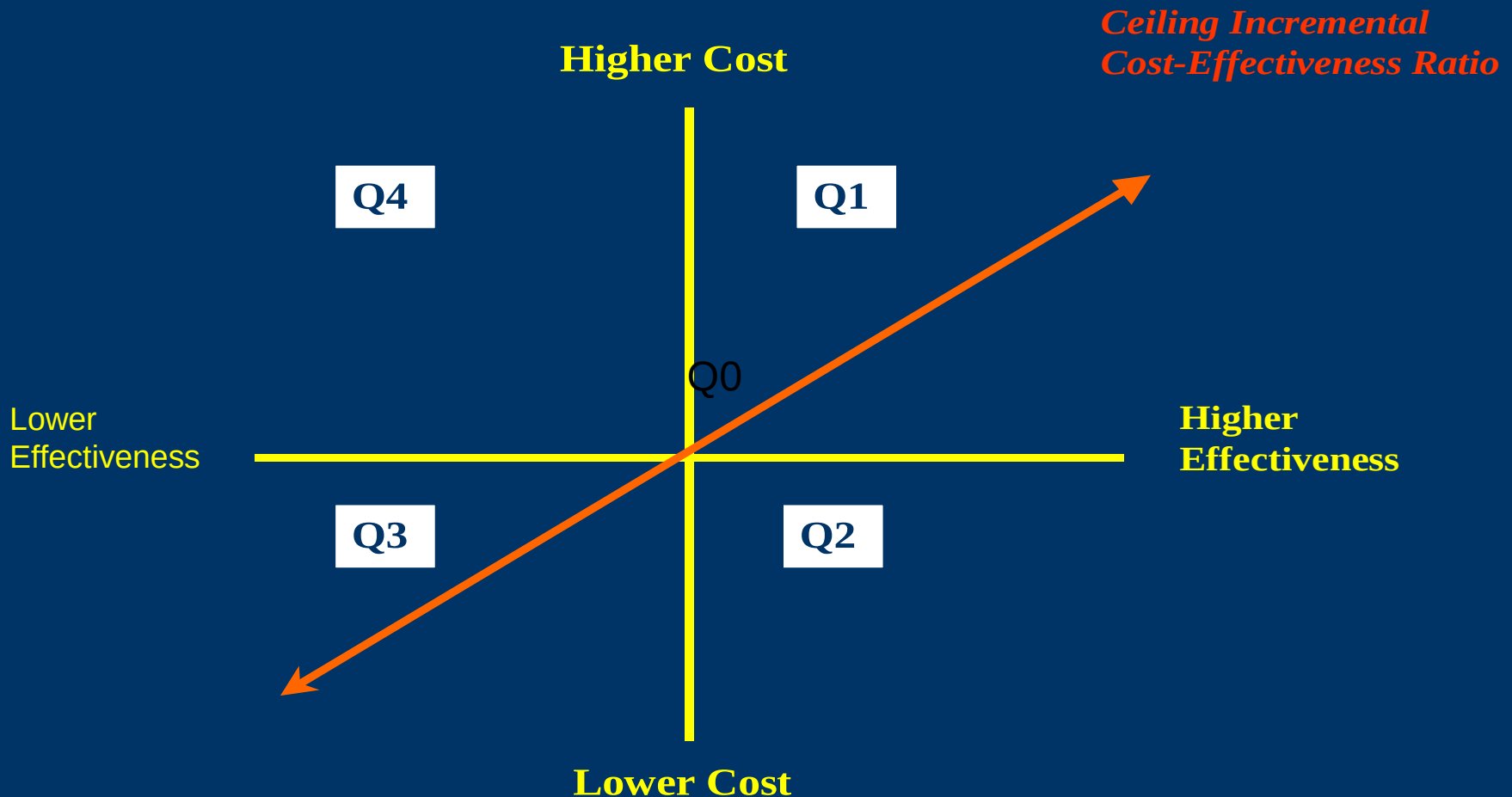
# 1. Cost Minimization Analysis

- **Costs** of intervention are compared
- Easiest because **outcomes are assumed to be equivalent** or the quality of both outcomes is assumed to be equal  
e.g. Promotion of generic prescribing

## 2. Cost-Effectiveness Analysis

- Summarizes **health benefits and resources** used by
  - **competing healthcare programs**
- involves comparing **treatment alternatives with different safety and efficacy** profiles
- **Cost** is measured in **monetary terms**, and

# The Cost-Effectiveness Plane



A is dominated by Q2  
Q4 is dominated by A

Decision on A vs Q1 and A vs Q3 depends  
on ICER and Budget

### 3. Cost-Utility Analysis

- Can compare **cost**, **quality**, and the **quantity of patient-years**
- Cost is measured in **dollars**, and
- Therapeutic outcome is measured in **patient-weighted utilities** rather than in **physical units**

## 4. Cost-Benefit Analysis

- Both resources and health consequences **are both valued in monetary units**, permitting costs and consequences to be compared against one another
- Health effects assigned a monetary value using
  - Willingness-to-pay (WTP) survey method
  - Human Capital (productivity gain/loss) method
- **Decision index** expressed as:
  - Net value: benefits minus costs-**most convenient**
  - Benefit to cost ratio

# Steps in PE evaluation

## **The main steps in PE evaluation are:**

- Defining the economic question and perspective
- Determining the alternatives to be evaluated
- Choosing the evaluation design
- Identifying, measuring and valuing the costs
- Identifying, measuring and valuing the benefits
- Adjusting costs and benefits for the differential timing
- Measuring incremental costs and benefits
- Putting the costs and benefits together
- Testing the sensitivity of the results
- Presenting the results

# Summary

| Methodology                       | Cost Measurement Unit   | Outcome Measurement Unit                                                            |
|-----------------------------------|-------------------------|-------------------------------------------------------------------------------------|
| Cost Minimization Analysis (CMA)  | Dollars /Monetary Units | Assumed to be equivalent in comparable groups                                       |
| Cost Effectiveness Analysis (CEA) | Dollars /Monetary Units | Natural units (life-years gained, SFDs, mm Hg blood pressure, mMol/L blood glucose) |
| Cost-Benefit Analysis (CBA)       | Dollars /Monetary Units | Dollars /Monetary Units                                                             |
| Cost Utility Analysis (CUA)       | Dollars /Monetary Units | Quality-adjusted life-year (QALY) or other utilities                                |

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